

Aaditya Vikram Saravana Bhavan

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Education

Carnegie Mellon University

Pittsburgh, PA

Master of Computational Data Science (MCDS), GPA: 4.00/4.00

Dec 2026

Relevant Courses: Introduction to Deep Learning (PhD), Introduction to Machine Learning (MS), Foundations of Computational Data Science (MS), Cloud Computing (MS), Multimodal Machine Learning (MS)

PES University

Bangalore, India

B.Tech in Computer Science and Engineering, GPA: 9.42/10.00

May 2024

Relevant Courses: Data Structures and Algorithms, Computer Networks, Data Analytics, Big Data

Experience

Amazon Web Services

Seattle, WA

Software Development Engineer Intern, Amazon RDS

May 2026 – Aug 2026

- Engineered an idempotent Java-based RDS control-plane workflow to integrate and propagate RSA-2048, RSA-4096, and ECC-384 regional CA certificates across 3 upcoming AWS Regions, enabling TLS-authenticated connectivity to RDS database instances
- Built 2 AWS Lambda functions and extended AWS CDK and IAM infrastructure to securely transfer certificate artifacts from a tooling-account S3 bucket to a global publication account, reducing public trust-store publication from weeks to days
- Automated internal trust-store onboarding across 2 repositories, 4 branches, and 12 configuration files by retrieving regional certificates, generating code reviews, and integrating failure ticketing into region-build automation

Cisco

Chennai, India

Software Development Engineer at Catalyst Center

Sep 2024 – Jul 2025

- Led development of a Go-based telemetry middleware, validating pipelines via AWS Athena for 10K+ customers; developed Webex bots (Node.js) for real-time alerts, reducing MTTR by 40%
- Improved reliability by 35% by resolving 50+ issues and standardizing error handling across 20+ microservices

Zscaler

Bangalore, India

Software Development Engineer Intern

Feb 2024 – Aug 2024

- Reduced data size by 40% and transfer latency by 50% using Protocol Buffers; optimized PostgreSQL and PL/pgSQL workloads
- Secured RabbitMQ communication, automated server workflows with Python, and achieved 95% unit test coverage

Software Development Engineer Intern

Jun 2023 – Aug 2023

- Improved system-level C serialization pipelines and reduced code duplication using X-Macros

Academic Projects

High-Throughput Twitter Recommendation Service

Pittsburgh, PA

Carnegie Mellon University, Cloud Computing

Jan 2026 – Apr 2026

- Built a Twitter recommendation service over 1 TB+ of data using PySpark, Go, Amazon EKS, Kubernetes, and Aurora MySQL, achieving 6,000 peak RPS and 5,000 sustained RPS at 99.7% correctness
- Increased throughput from 200 to 5,000 sustained RPS through schema denormalization, indexed Aurora lookups, horizontal pod scaling, connection-pool tuning, and concurrent authentication and database queries

MyTorch

Pittsburgh, PA

Carnegie Mellon University, Intro to Deep Learning

Aug 2025 – Dec 2025

- Built a PyTorch like deep learning library from scratch, implementing forward/backward propagation, autograd style training, and models including MLPs, CNNs, RNNs (GRU/LSTM), and Transformers

GAN-Based Multispectral Image Dehazing

Bangalore, India

PES University

Jan 2023 – Apr 2024

- Developed a GAN-based model in TensorFlow/Python for multispectral image dehazing, outperforming existing methods on the SHIA dataset
- Awarded Best Paper (ISMSI 2024) for advancing state of the art image dehazing techniques

Yet Another Kafka

Bangalore, India

PES University

Nov 2022 – Jan 2023

- Built a Kafka-like distributed messaging system using socket programming with producer-consumer architecture
- Implemented multi-broker replication with ZooKeeper recovery, enabling fault tolerance and client subscriptions

Publications

- Enhancing Multispectral Vision: A GAN-Based Dehazing Framework, ISMSI 2024
- A Pugnacious Comparative Study of Data Analysis Techniques for Wine Quality, I2CT 2024
- Android Malware Detection: A Comprehensive Review, Research Advances in Network Technologies 2024

Skills

Languages: Go, C/C++, Python, Java, SQL; **LeetCode Knight (Top 2.75%)**

Cloud & Systems: AWS, Distributed Systems, Microservices, REST APIs, Linux, Docker, Kubernetes, AWS CDK

Databases & Data: Aurora MySQL, PostgreSQL, MongoDB, PySpark, AWS Athena

Machine Learning: PyTorch, TensorFlow, scikit-learn, Transformers, Computer Vision